STUMPRIDER · DESIGN DECISION RECORD

StumpRider decisions of 2026-10-25 (heel limit, concept, crutch, canoe eligibility, hook head)

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	Amish's decisions O6 to O9 of 2026-10-25 recorded and carried out; O10 approved but stopped while carrying it through

0006: Decisions of 2026-10-25

- **Date:** 2026-10-25
- **Status:** accepted (O6 to O9); O10 returned to Amish

Context

The kit 1.9.1 upgrade and the independent review of the calculation note left five open decisions in SMR-DEC-001 (O6 to O10), each with a recommendation, listed in a report to Amish. Amish Chadha (owner), 2026-10-25 12:43 CT: "i approve of your recommendations". Every one carried a recommendation, so each is recorded as decided with the option recommended, and carried through. One of them, O10, did not survive being carried through and is back with Amish (below).

> **Safety:** O6 and O9 set how the tool is used from a canoe. Both keep the conservative limits: the 2.0 mm pin, the 5 deg static heel limit, the 7 m and three-crew rule, kneeling, slackening the line and life jackets. Nothing here puts a person in the water.

Decision

1. **O6, option A (heel limit for a quick pull, PR-003).** The 2.0 mm pin stays. 5 deg is the static heel limit. "Lift steadily, never jerk" joins the operating rules (build plan stop S4a) and the pictorial tag (PR-013). The TRL 4 moored heel test records the roll period and the peak heel under a quick pull; a 1.7 mm pin (185 to 251 N, peak about 4.9 deg) only if that test shows a real exceedance. No cost.
2. **O7, option A (concept selection).** Option A, the rider ring on a shear-pin pole, with option B's jigging already in it. `concept_chosen: A`; SMR-DDR-005 accepted. The eligibility rule offered beside the options was not part of the recommendation and is not adopted.
3. **O8, option A (crutch).** Each canoe keeps its own crutch; only the jigging weight is shared by five canoes. The carried mass is unchanged (bamboo variant 3.95 kg, PR-007); the bamboo local variant costs USD 66.90 a kit (estimate; USD 54.90 when the crutch was shared). The jigging weight's galvanising share is taken as half the USD 6 the crutch and weight carried together (estimate). `bom/bom-local-variant.csv` lines 15 to 18 updated.
4. **O9, option A (canoe eligibility, PR-003).** The 7 m, three-crew rule stays as a screen, and the TRL 4 heel test adds a measured criterion: displacement times GM of at least about 260 kg m, the value that keeps the static heel at 5 deg, with a margin set from the test. PR-003 restated.
5. **O10, option C (hook head tip leg), approved but stopped.** The recommendation was a small gusset outside the bend, about 0.02 kg, to bring the tip-leg case (factor 1.58 at the bend) over the 2 x rule of PR-012. Carrying it through, the calculation found that the bight on the tip leg puts the same moment, the load times 60 mm, into the whole straight shank between the bend and the flat, where the factor is 1.87 (SMR-CAL-001 [S13a]). A gusset at the bend does not reach that section; a gusset that runs to the flat (about 0.06 to 0.10 kg, estimate) takes the bamboo variant past PR-007's 4.0 kg, which is the trade the recommendation was chosen to avoid. Under the brief's rule the change was not made; O10 is open again in SMR-DEC-001 with the options restated.

Consequences

- **Requirements.** PR-003 restated (static limit, steady-lift rule, measured criterion); PR-010 not met at USD 66.90; PR-012 still at risk (hook head tip leg 1.58 at the bend, 1.87 in the shank; bamboo pole 1.49); PR-013 tag carries "lift steadily, never jerk".
- **STEM check.** S7 restated and plausible (to confirm at TRL 4); S3b stays failing with the new finding; 15 within known science, 11 plausible, 1 failing.
- **Design.** No change to the model, drawings or BOM of the aluminium prototype; `budget_usd` unchanged. Value-engineering target: USD 2,000. Estimated cost of the constructable design: USD 299.50 for the prototype run (unchanged).
- **Still open.** O10.